

Predicting the Second Gift

Which first-time donors will give again, and which ones a small nonprofit should spend staff time on.

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DISCLOSURE, SAID FIRST

One of us founded GoGood Technologies, a donor-engagement platform whose customers are the stakeholder in this talk. That is why we chose the problem. Weigh what we say accordingly.

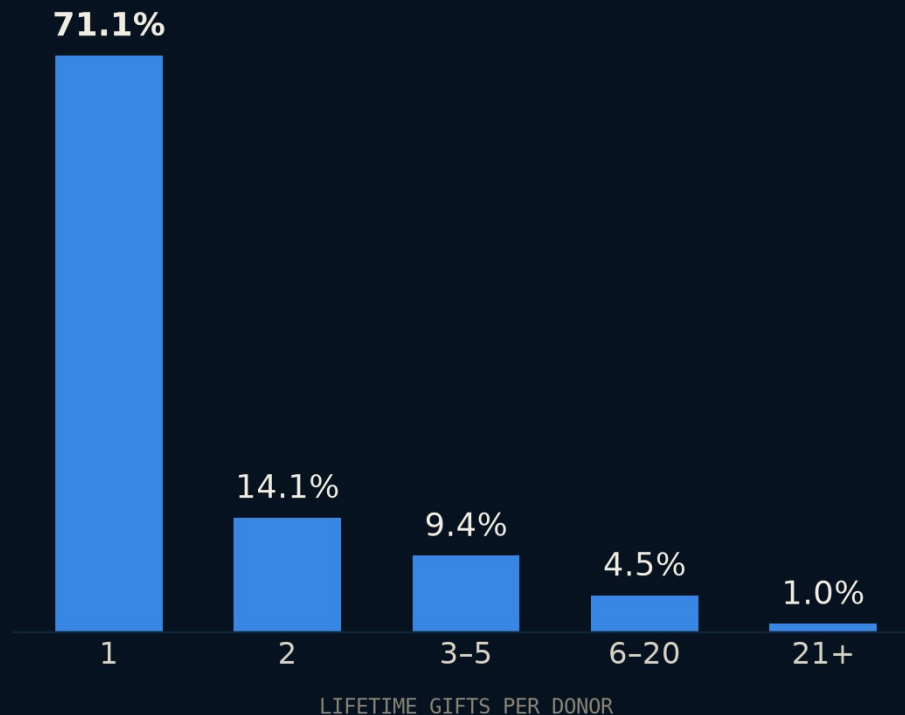
71%

of Donors Choose donors gave once
and never again.

3.5 million donors, seventeen years. The second gift is where retention is won or lost, and small organizations have the least capacity to work it.

OUR STAKEHOLDER

A development lead at a ~\$500K nonprofit, no data staff. Each month she can personally follow up with a fraction of last month's new donors. She builds that list by hand, from recency and gift size.



Four decisions turn eleven million rows into one label. Each could have gone the other way.

11.4M

DONATIONS

3.28M

LABELLED DONORS

15.8%

REPEAT WITHIN 12 MO

ICPSR 37898, 2002–2019. Train through 2016, hold out 2017–18: a random split would let the model see the future.

1 Same-month repeats don't count

One checkout can fund several classrooms. Two gifts in month one are one event, not a return. She cannot steward back someone who never left.

2 Twelve whole months

M+1 to M+12. Dates are month-level, so no finer definition exists and none is claimed.

3 Unclosed windows: dropped, not zeroed

A 2019 donor has not had a year to come back. Labelling them 0 teaches the model that recent donors never return.

4 Refunds are not gifts

The codebook lists a minimum amount of –\$15. A reversal cannot open a cohort or count as a return. 151 rows, excluded.

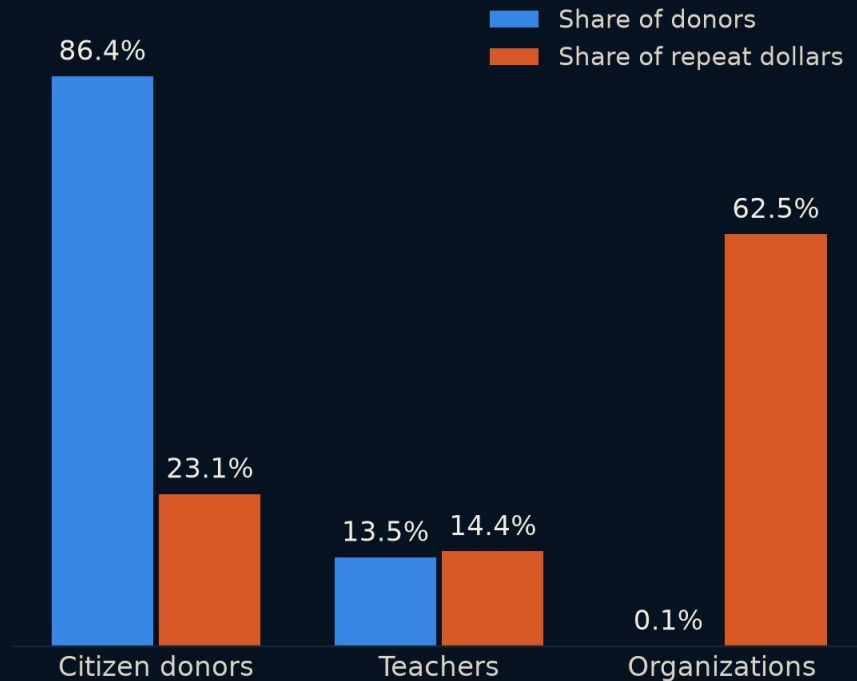
708 organizations hold 62% of every repeat dollar.

Our first result found 80% of all repeat dollars at 10% capacity with the dumbest possible rule: rank by gift size. It looked like a triumph.

It was measuring corporate matching programs. The largest made 66,348 donations in its twelve-month window. Any ranker wins the pooled contest by finding them — and she already knows who they are.

THE DECISION

Score citizen donors only — 86% of people, the ones she actually calls. One line in config, reversible, both numbers kept.



What predicts a return

Gift size finds dollars; [feature] finds people. Here is what moved and what did not.

Structure to keep: one chart of importance or lift, one sentence per feature that mattered, one on what did not.

Text is masked by ICPSR — no embeddings. Tie back to slide 4.

DROP HERE

Feature importance or lift table — one chart

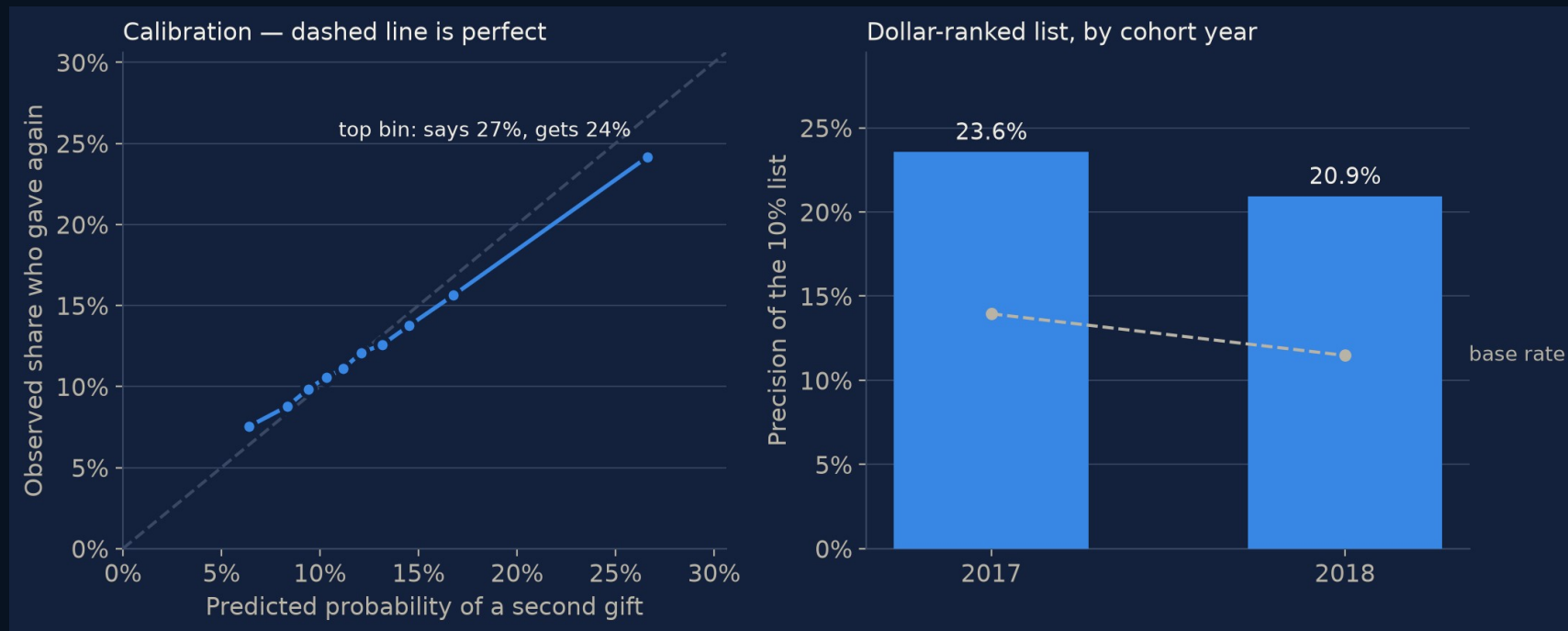
At her capacity, dollars identified per contact

Citizen donors · top 10% of each month's new donors · 2017–18 holdout · value identified, never “caused”

RANKING	PRECISION	RECALL	VALUE FOUND	\$ / CONTACT
Our model — probability × amount score	22.1%	17.6%	58.3%	\$120
First gift size — the honest baseline	19.7%	15.7%	56.4%	\$116
First-month gift count	20.4%	16.2%	41.4%	\$85
Random	12.7%	10.1%	9.1%	\$19
Contact everyone	12.6%	100%	100%	\$21

30 paired donor bootstrap samples: $+\$3.80 \pm 0.51$ per contact over gift size (± 1 standard error). The 3.3% gain is modest; gift size already carries most of the dollar signal. The baseline is fair: on a first-gift cohort, RFM collapses to gift size, and it is what she does by hand.

The gain holds, but the dollar list favors large gifts



Under-\$50 donors: 52.4% of the holdout, only 0.55% of the dollar-ranked list (14.4% if ranked by probability).

What she does on Monday

3,013

SELECTED · DEC 2018

\$25

PER CALL ·
PLACEHOLDER

\$7.75M

VALUE LESS ASSUMED
COST

No API

EXTERNAL MODEL CALLS

The model's list contains \$9.79M of subsequent giving. Subtracting \$2.04M of hypothetical contact costs leaves \$7.75M, versus \$7.44M for gift-size ranking. Neither balance measures outreach profit: we do not know how much extra giving a call causes.

COST AT PRODUCTION SCALE

Scoring runs locally without API fees. Staff outreach, setup and maintenance still cost money. Replace the \$25 contact-cost assumption with evidence from practice.

What we did not find, and what we are not claiming

1

Outreach impact is unknown

The records show who gave again, not what a call changed. Dollars identified minus assumed contact cost is not measured profit.

2

Thank-you packet: dropped

The codebook gives no timing. The flag may record something that happened after the second gift. Albert pre-approved dropping it.

3

Transfer is untested

We tested 2017–18 DonorsChoose cohorts. Performance for today's donors or another nonprofit has not been established.

Next: validate locally, measure real contact costs, and test randomized outreach before scaling.

Pilot the model-ranked list against gift-size ranking. The historical top 10% contained 58% of subsequent giving; test whether outreach adds donations before scaling.